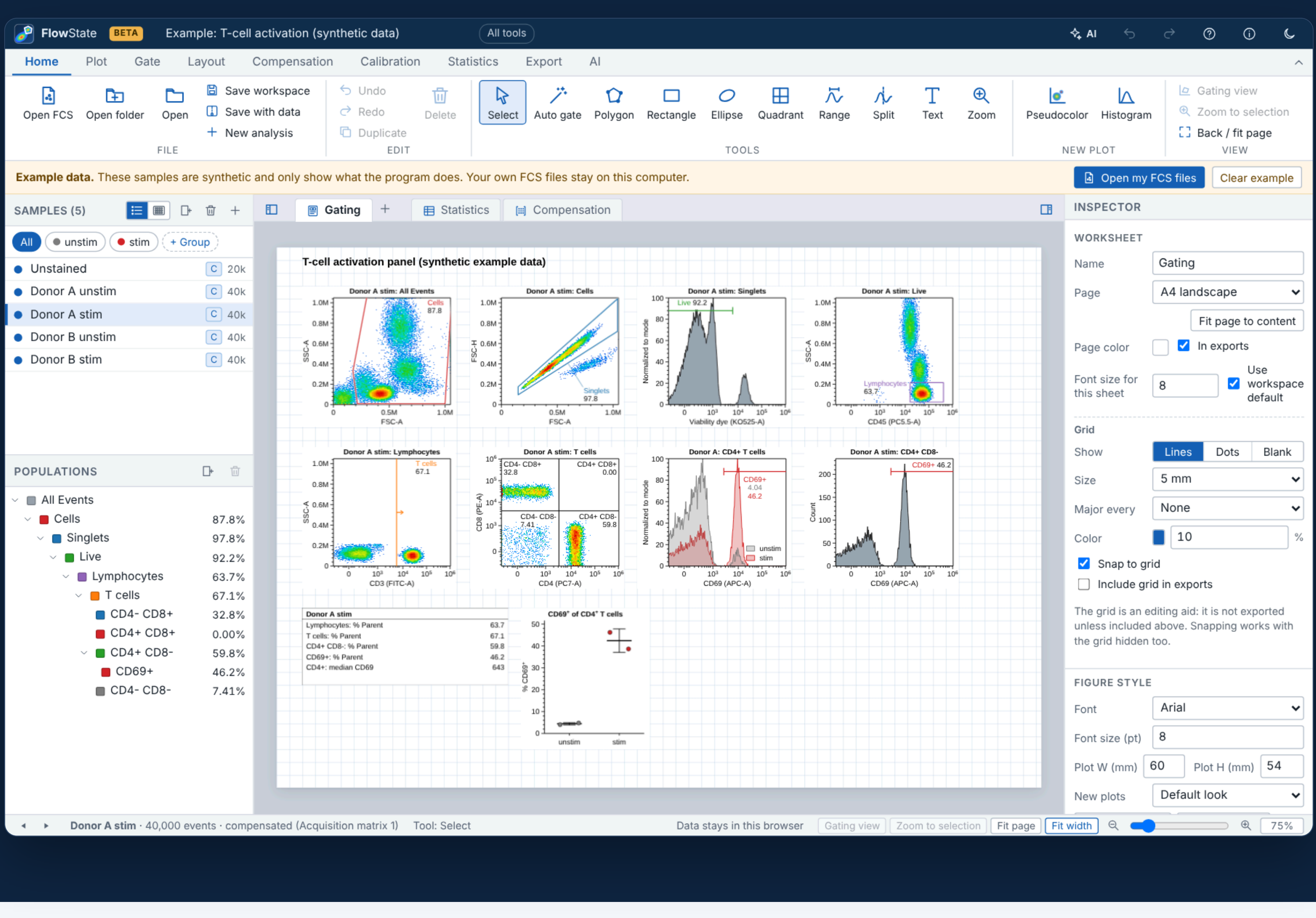
FlowState

Free and open source · Beta

Flow cytometry analysis and publication-ready figures, right in your browser.

Gating, statistics, compensation, spectral unmixing, EV size and MESF calibration, FlowJo import. No installation and no upload: your FCS files never leave your computer.

FCS 2.0 to 3.2 and Beckman LMD | BD · Beckman · Cytek · Sony · Thermo · Miltenyi · Agilent | Works offline | Autosaves in the browser



Example data. These samples are synthetic and only show what the program does. Your own FCS files stay on this computer.

Unstained	C	20k
Donor A unstim	C	40k
Donor A stim	C	40k
Donor B unstim	C	40k
Donor B stim	C	40k

POPULATIONS

All Events	
Cells	87.8%
Singlets	97.8%
Live	92.2%
Lymphocytes	63.7%
T cells	67.1%
CD4- CD8+	32.8%
CD4+ CD8+	0.00%
CD4+ CD8-	59.8%
CD69+	46.2%
CD4- CD8-	7.41%

Donor A stim	
Lymphocytes: % Parent	63.7
T cells: % Parent	67.1
CD4+ CD8-: % Parent	59.8
CD69+: % Parent	46.2
CD4+: median CD69	643

Name: Gating
Page: A4 landscape
Page color: ☐ ☑ In exports
Font size for this sheet: 8 ☑ Use workspace default
Show: Lines / Dots / Blank
Size: 5 mm
Major every: None
Color: 10 %
☑ Snap to grid
☐ Include grid in exports

The grid is an editing aid: it is not exported unless included above. Snapping works with the grid hidden too.

Font: Arial
Font size (pt): 8
Plot W (mm): 60 Plot H (mm): 54
New plots: Default look

Donor A stim · 40,000 events · compensated (Acquisition matrix 1) Tool: Select Data stays in this browser

Get started in three steps

1 Open FlowState

Download flowstate-standalone.html and double-click it (Chrome, Edge, Firefox or Safari). Nothing to install.

2 Drop your files

Drag FCS or LMD files, a FlowJo .wsp, a Gating-ML file or a saved FlowState workspace onto the window.

3 Gate, measure, export

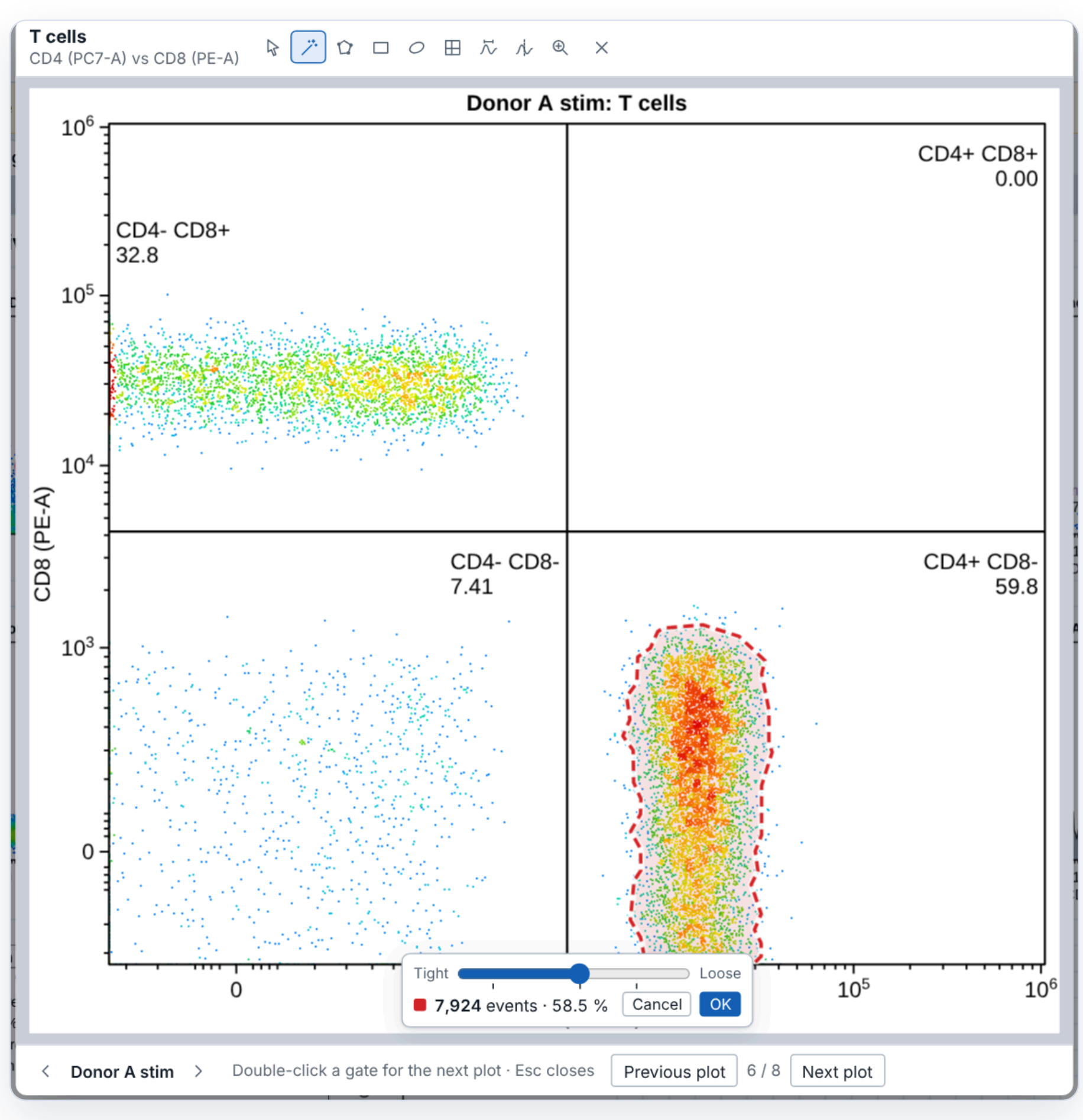
Draw or auto-gate, add statistics, lay out the figure and export PDF, SVG, PNG or TIFF. Press ? for all shortcuts.

01 · GATING

Gate quickly, count exactly

Draw gates on any plot, or let FlowState find the population for you. Counts are checked against FlowKit and FlowJo.

- Polygon, rectangle, ellipse, quadrant, range, split and Boolean (AND, OR, NOT) gates
- Auto gate: click a cluster or drag a box, tune it from tight to loose, then OK
- Gating view: open any plot large with Enter or a double-click
- Gate trees per group or per sample, copy gates between groups, adjust a gate for one sample only
- Full undo history, autosave, and workspaces you can reopen anywhere



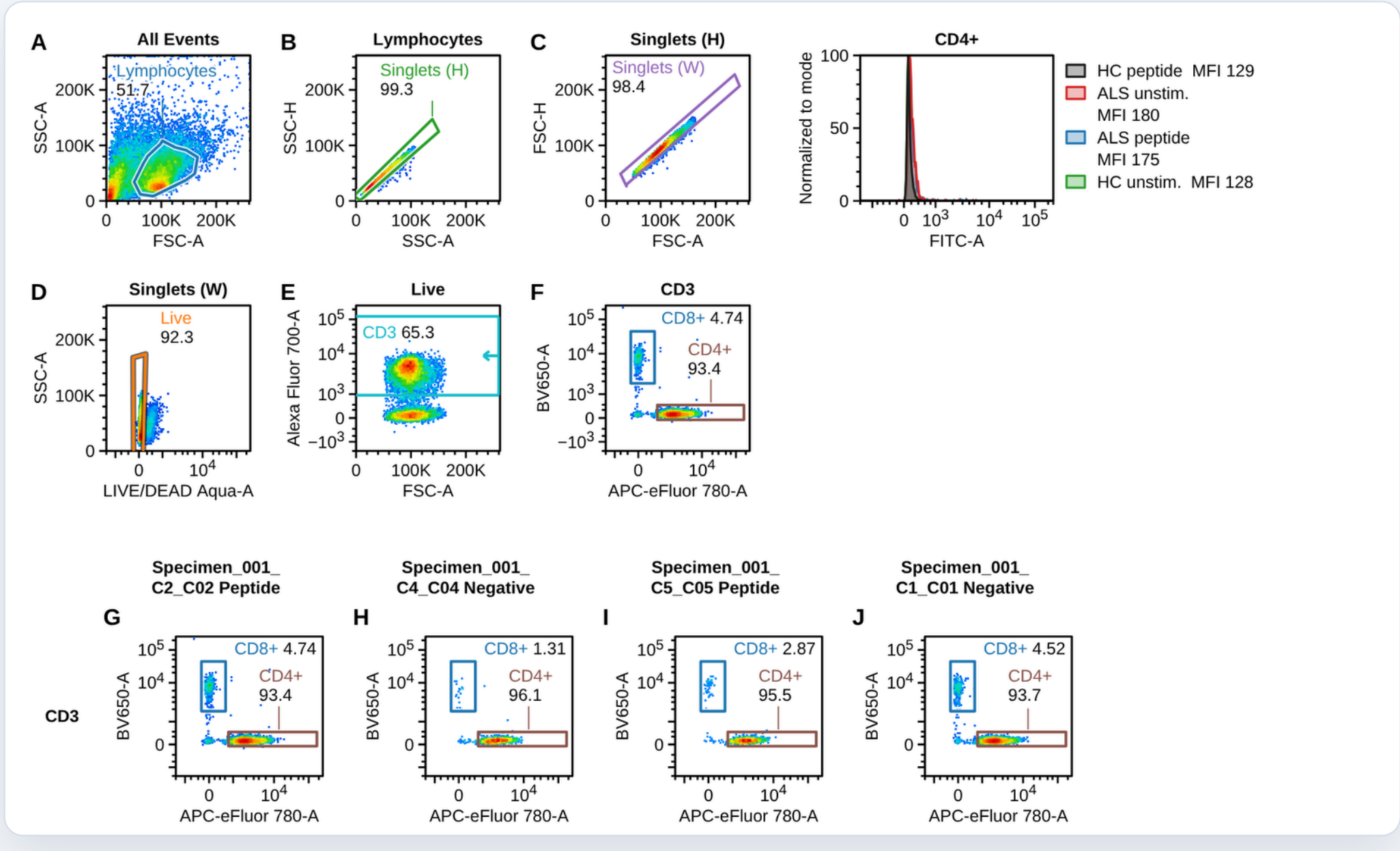
Auto gate on CD4 vs CD8: the boundary comes from the event density.

02 · FIGURES

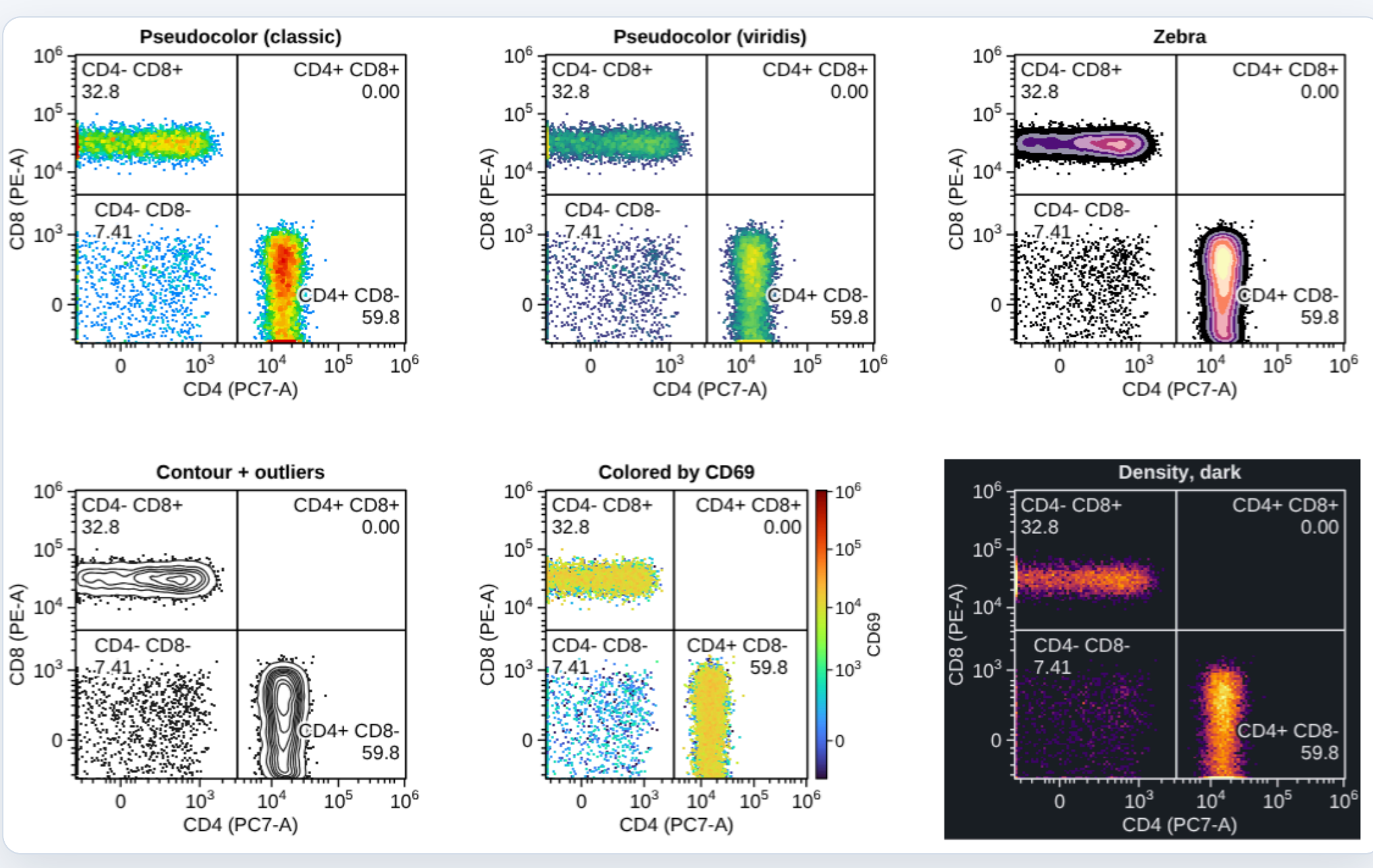
Figures ready for the journal

Lay out panels on a page in millimetres and export exactly that size. Gate labels and legends place themselves where they don't cover data.

- Journal widths (85, 120, 180 mm), A4 and 16:9 slide pages, snapping and equal gaps
- Repeat a plot for every sample and add panel letters in one click
- PDF with embedded fonts, SVG with editable text, PNG and TIFF up to 1200 dpi
- A gating-strategy sheet built from your gate tree



Exported figure from a public FlowJo ICS workspace (Zenodo 16754553).



One plot, six looks.

Every look you need

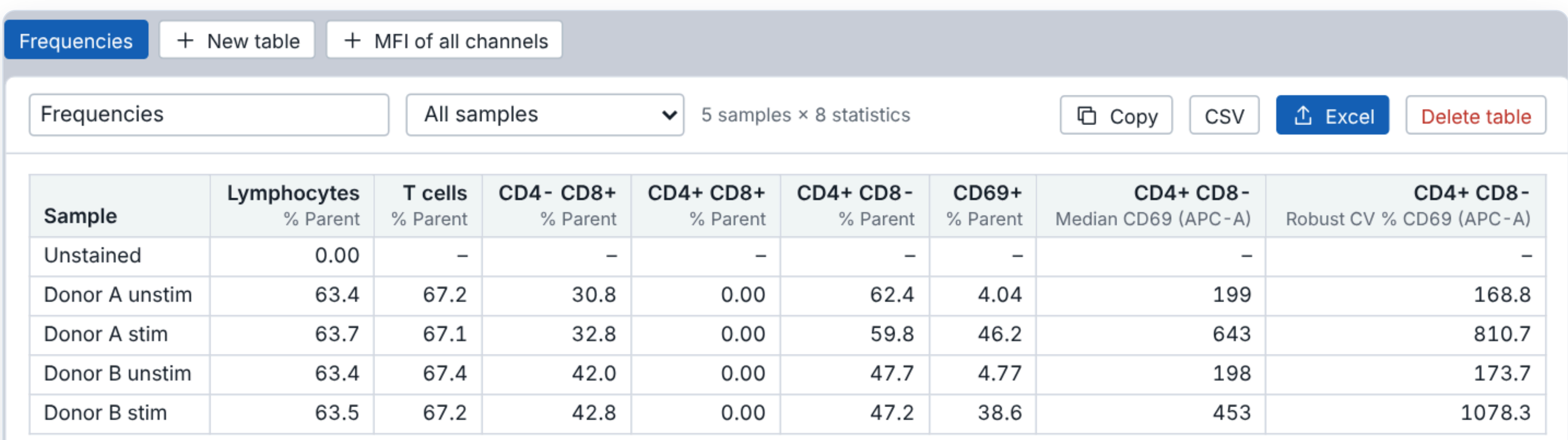
- 20 colormaps including viridis and magma: pseudocolor, zebra, contour, density and dot plots
- Color by a third parameter with a color bar, backgating, histogram overlays
- Style presets for print, dark slides and minimal axes
- No smoothing unless you switch it on; statistics are never smoothed

03 · STATISTICS

Statistics tables in seconds

Choose populations and statistics once; the table fills in for every sample and updates when you move a gate.

- % of parent, of total or of any ancestor, counts and events/µL
- Median, mean, geometric mean, SD, CV, robust CV, percentiles, mode
- MFI of all channels with one button; Excel, CSV and copy to clipboard
- Bar and dot charts per sample or group, with error bars



Frequencies

5 samples × 8 statistics

Sample	Lymphocytes % Parent	T cells % Parent	CD4- CD8+ % Parent	CD4+ CD8+ % Parent	CD4+ CD8- % Parent	CD69+ % Parent	CD4+ CD8- Median CD69 (APC-A)	CD4+ CD8- Robust CV % CD69 (APC-A)
Unstained	0.00	–	–	–	–	–	–	–
Donor A unstim	63.4	67.2	30.8	0.00	62.4	4.04	199	168.8
Donor A stim	63.7	67.1	32.8	0.00	59.8	46.2	643	810.7
Donor B unstim	63.4	67.4	42.0	0.00	47.7	4.77	198	173.7
Donor B stim	63.5	67.2	42.8	0.00	47.2	38.6	453	1078.3

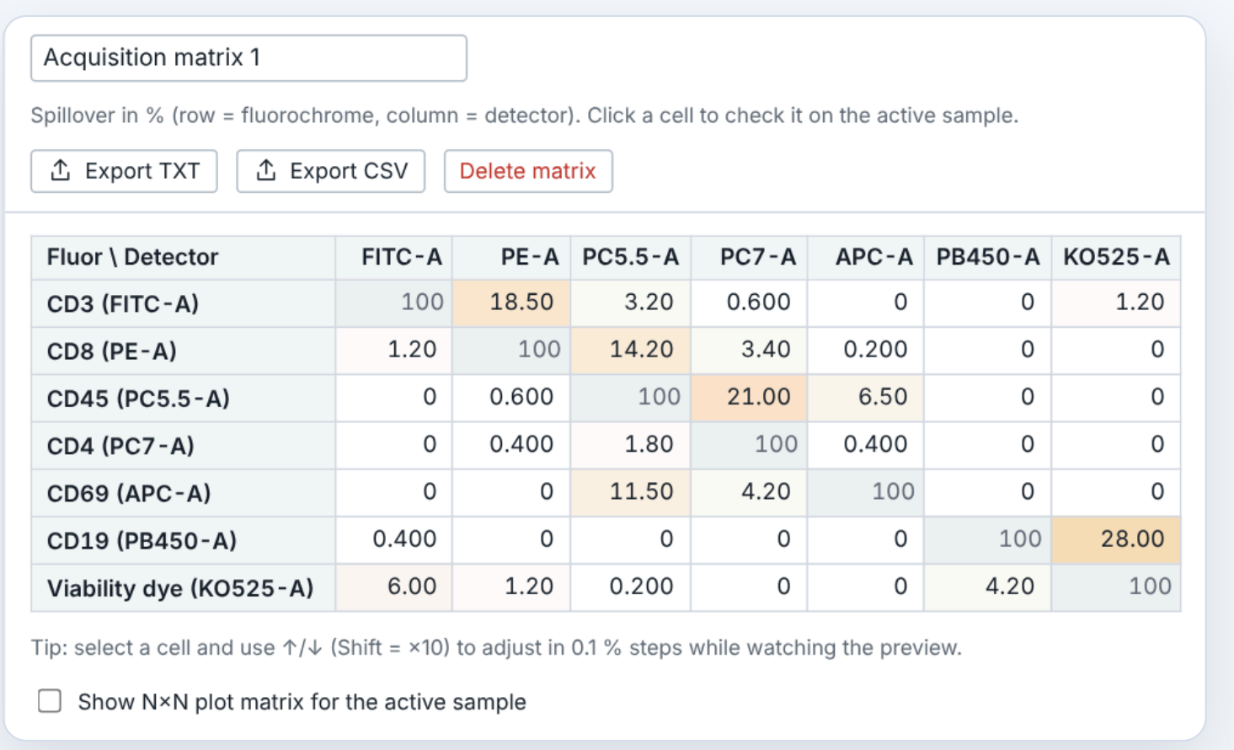
A frequency and MFI table for all samples (built-in example data).

04 · COMPENSATION AND SPECTRAL

Compensation and spectral unmixing

Use the cytometer's matrix, fine-tune it with a live preview or compute it from single stains. Unmix spectral panels with quality control built in.

- Matrix editor with live preview and an N×N plot matrix
- Import CytExpert .comp, FlowJo and text matrices; export TXT and CSV
- Spectral unmixing (OLS or WLS) with autofluorescence as its own channel
- Similarity, complexity, spreading and leakage checks; Aurora controls agree with Cytek's unmixing (r ≥ 0.999)



Acquisition matrix 1

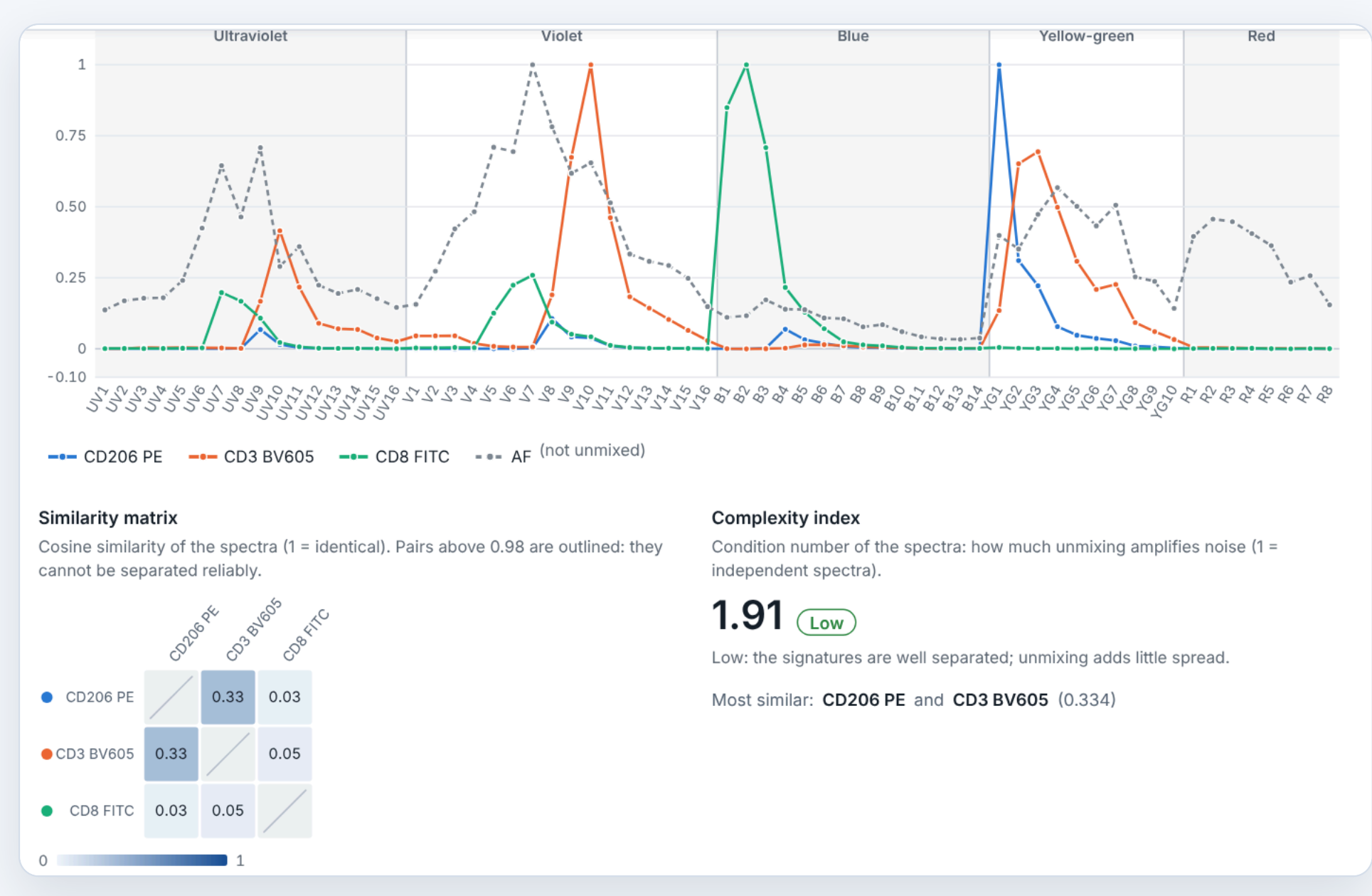
Spillover in % (row = fluorochrome, column = detector). Click a cell to check it on the active sample.

Fluor \ Detector	FITC-A	PE-A	PC5.5-A	PC7-A	APC-A	PB450-A	KO525-A
CD3 (FITC-A)	100	18.50	3.20	0.600	0	0	1.20
CD8 (PE-A)	1.20	100	14.20	3.40	0.200	0	0
CD45 (PC5.5-A)	0	0.600	100	21.00	6.50	0	0
CD4 (PC7-A)	0	0.400	1.80	100	0.400	0	0
CD69 (APC-A)	0	0	11.50	4.20	100	0	0
CD19 (PB450-A)	0.400	0	0	0	0	100	28.00
Viability dye (KO525-A)	6.00	1.20	0.200	0	0	4.20	100

Tip: select a cell and use ↑/↓ (Shift = ×10) to adjust in 0.1 % steps while watching the preview.

☐ Show N×N plot matrix for the active sample

Spillover matrix in %, editable with a live preview.



Similarity matrix

Cosine similarity of the spectra (1 = identical). Pairs above 0.98 are outlined: they cannot be separated reliably.

	CD206 PE	CD3 BV605	CD8 FITC
CD206 PE		0.33	0.03
CD3 BV605	0.33		0.05
CD8 FITC	0.03	0.05	

Complexity index

Condition number of the spectra: how much unmixing amplifies noise (1 = independent spectra).

1.91 Low

Low: the signatures are well separated; unmixing adds little spread.

Most similar: **CD206 PE** and **CD3 BV605** (0.334)

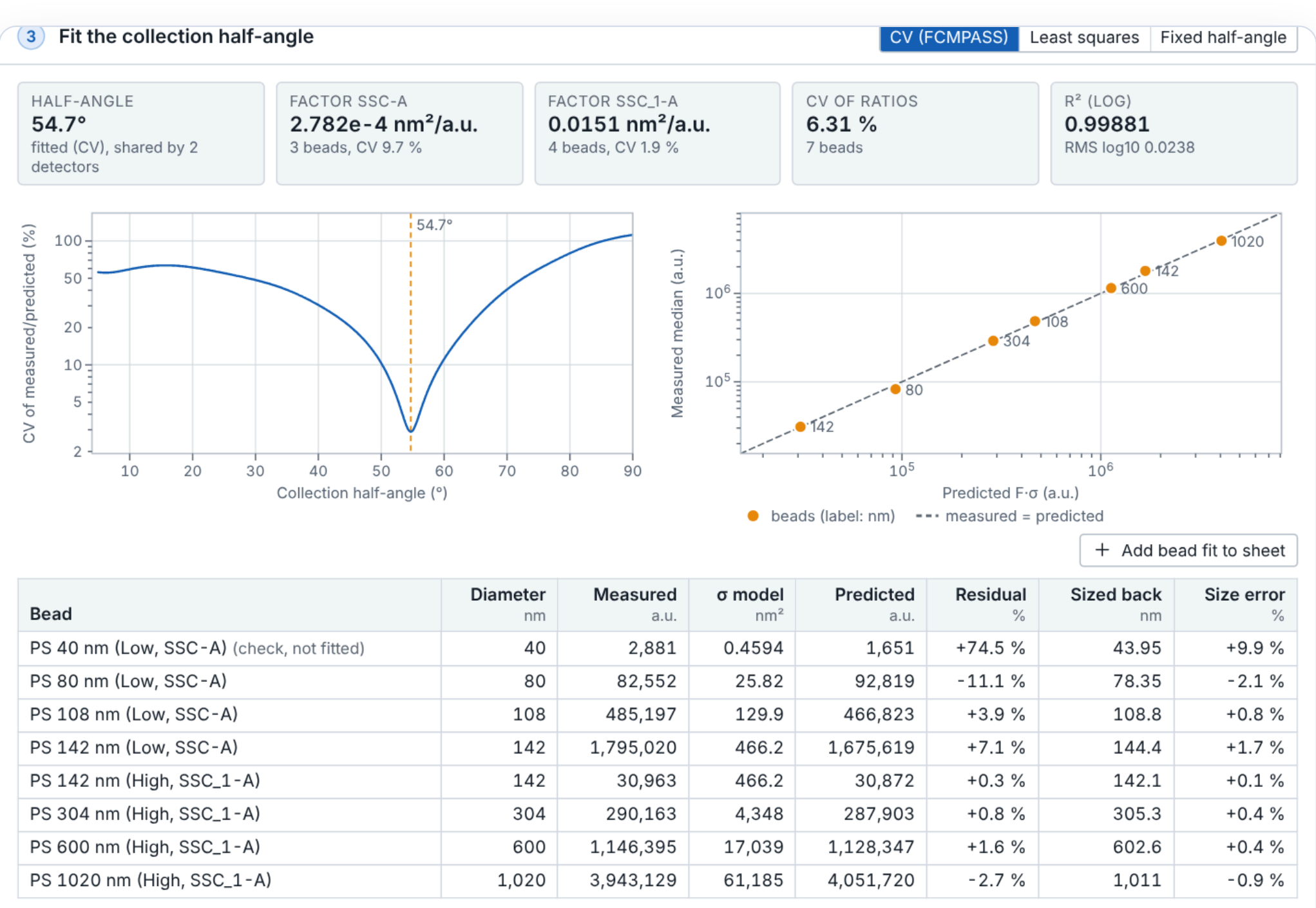
Reference spectra and similarity matrix from Cytek Aurora controls (public data).

05 · NANOSCALE

EV sizing and MESF calibration

Turn side scatter into particle diameter with Mie theory, as in FCMPASS, and fluorescence into MESF or ERF units.

- nanoViS Low and High presets for the CytoFLEX nano (VSSC1 and VSSC2)
- Fits the collection half-angle and sizes the beads back as a check
- New channels: diameter (nm) and scattering cross section (nm²)
- Saturation warnings, valid size range and a MIFlowCyt-EV report



HALF-ANGLE 54.7° fitted (CV), shared by 2 detectors
FACTOR SSC-A 2.782e-4 nm²/a.u. 3 beads, CV 9.7 %
FACTOR SSC_1-A 0.0151 nm²/a.u. 4 beads, CV 1.9 %
CV OF RATIOS 6.31 % 7 beads
R² (LOG) 0.99881 RMS log10 0.0238

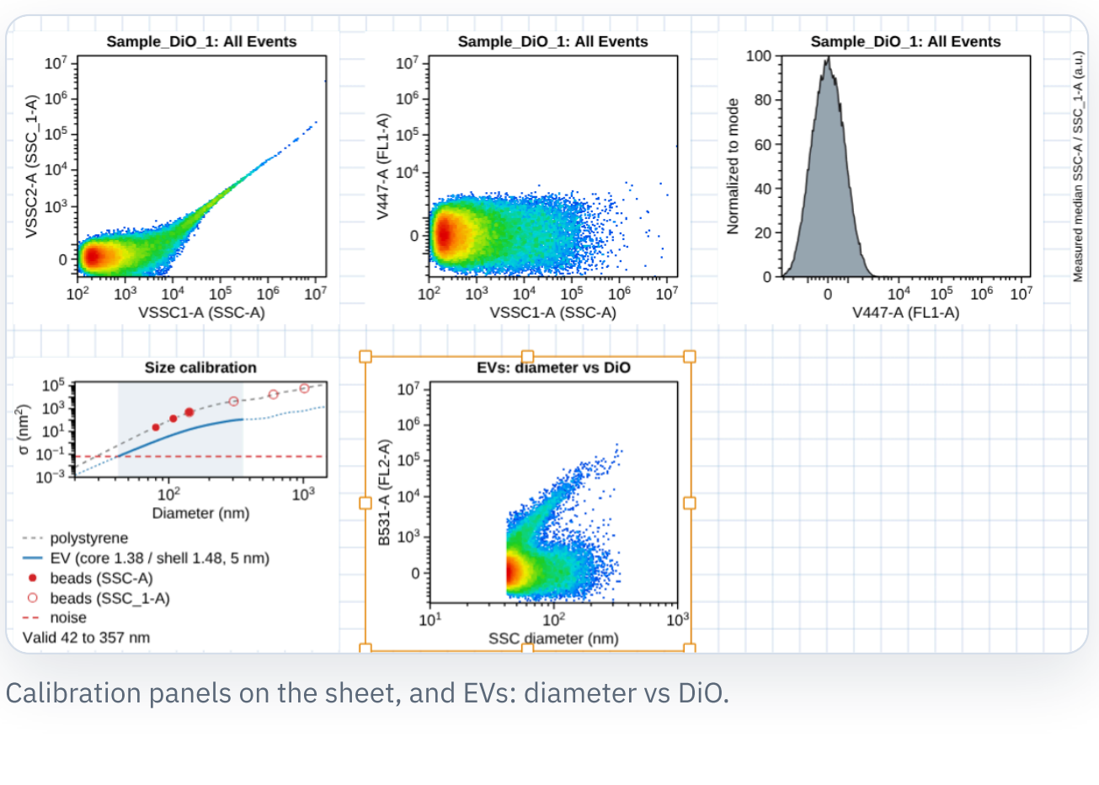
Bead	Diameter nm	Measured a.u.	σ model nm²	Predicted a.u.	Residual %	Sized back nm	Size error %
PS 40 nm (Low, SSC-A) (check, not fitted)	40	2,881	0.4594	1,651	+74.5 %	43.95	+9.9 %
PS 80 nm (Low, SSC-A)	80	82,552	25.82	92,819	-11.1 %	78.35	-2.1 %
PS 108 nm (Low, SSC-A)	108	485,197	129.9	466,823	+3.9 %	108.8	+0.8 %
PS 142 nm (Low, SSC-A)	142	1,795,020	466.2	1,675,619	+7.1 %	144.4	+1.7 %
PS 142 nm (High, SSC_1-A)	142	30,963	466.2	30,872	+0.3 %	142.1	+0.1 %
PS 304 nm (High, SSC_1-A)	304	290,163	4,348	287,903	+0.8 %	305.3	+0.4 %
PS 600 nm (High, SSC_1-A)	600	1,146,395	17,039	1,128,347	+1.6 %	602.6	+0.4 %
PS 1020 nm (High, SSC_1-A)	1,020	3,943,129	61,185	4,051,720	-2.7 %	1,011	-0.9 %

Sized back: the diameter at which each bead's own material reproduces its measured median under this calibration (should match the stated diameter within a few %).

Swept half-angles 5 to 90 ° in 0.1° steps

- nanoViS 40 nm bead: Beckman's CytoFLEX nano application note gives 44 nm for it (it sizes back to 43.9 nm here, as a check: it is not used in the fit); enter the certificate value of your lot if you have it.

Bead fit with real nanoViS files from a CytoFLEX nano (public data): half-angle 54.7°, beads sized back within a few %.



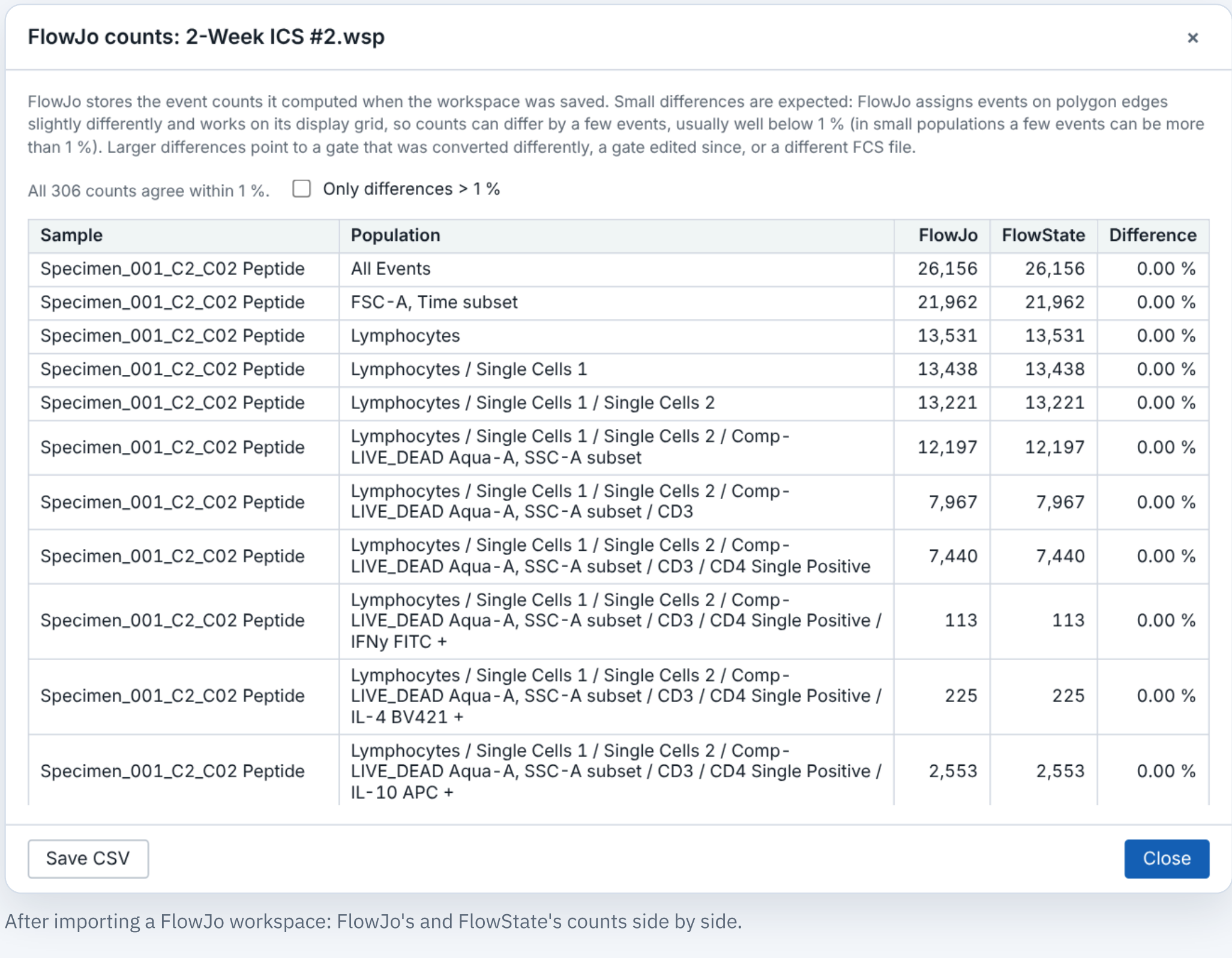
Calibration panels on the sheet, and EVs: diameter vs DiO.

06 · YOUR OTHER TOOLS

Works with FlowJo, CytExpert and FlowLogic

Bring existing analyses in and take results out in open formats.

- FlowJo workspaces (.wsp) with gates, groups and compensation, and a side-by-side count check
- Tested: 306 of 306 population counts identical to FlowJo on a public workspace
- Gating-ML 2.0 in and out, for FlowKit, R/flowCore and Cytobank
- Export populations as FCS 3.1 or CSV; FlowLogic-compatible matrices; CytExpert .comp import
- Checked on 166 public files from about 25 cytometer models



FlowJo counts: 2-Week ICS #2.wsp

FlowJo stores the event counts it computed when the workspace was saved. Small differences are expected: FlowJo assigns events on polygon edges slightly differently and works on its display grid, so counts can differ by a few events, usually well below 1 % (in small populations a few events can be more than 1 %). Larger differences point to a gate that was converted differently, a gate edited since, or a different FCS file.

All 306 counts agree within 1 %. ☐ Only differences > 1 %

Sample	Population	FlowJo	FlowState	Difference
Specimen_001_C2_C02 Peptide	All Events	26,156	26,156	0.00 %
Specimen_001_C2_C02 Peptide	FSC-A, Time subset	21,962	21,962	0.00 %
Specimen_001_C2_C02 Peptide	Lymphocytes	13,531	13,531	0.00 %
Specimen_001_C2_C02 Peptide	Lymphocytes / Single Cells 1	13,438	13,438	0.00 %
Specimen_001_C2_C02 Peptide	Lymphocytes / Single Cells 1 / Single Cells 2	13,221	13,221	0.00 %
Specimen_001_C2_C02 Peptide	Lymphocytes / Single Cells 1 / Single Cells 2 / Comp-LIVE_DEAD Aqua-A, SSC-A subset	12,197	12,197	0.00 %
Specimen_001_C2_C02 Peptide	Lymphocytes / Single Cells 1 / Single Cells 2 / Comp-LIVE_DEAD Aqua-A, SSC-A subset / CD3	7,967	7,967	0.00 %
Specimen_001_C2_C02 Peptide	Lymphocytes / Single Cells 1 / Single Cells 2 / Comp-LIVE_DEAD Aqua-A, SSC-A subset / CD3 / CD4 Single Positive	7,440	7,440	0.00 %
Specimen_001_C2_C02 Peptide	Lymphocytes / Single Cells 1 / Single Cells 2 / Comp-LIVE_DEAD Aqua-A, SSC-A subset / CD3 / CD4 Single Positive / IFNy FITC +	113	113	0.00 %
Specimen_001_C2_C02 Peptide	Lymphocytes / Single Cells 1 / Single Cells 2 / Comp-LIVE_DEAD Aqua-A, SSC-A subset / CD3 / CD4 Single Positive / IL-4 BV421 +	225	225	0.00 %
Specimen_001_C2_C02 Peptide	Lymphocytes / Single Cells 1 / Single Cells 2 / Comp-LIVE_DEAD Aqua-A, SSC-A subset / CD3 / CD4 Single Positive / IL-10 APC +	2,553	2,553	0.00 %

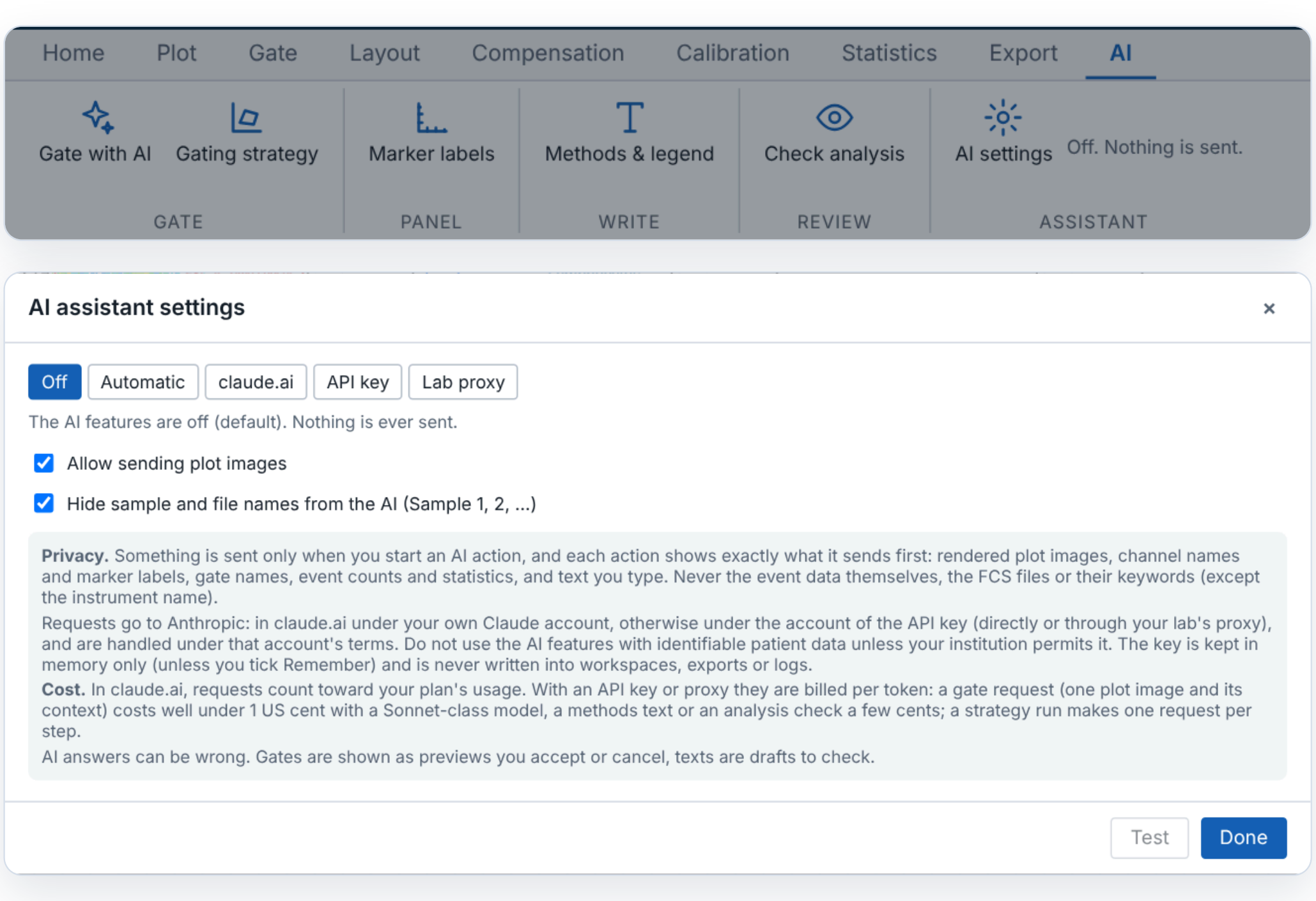
After importing a FlowJo workspace: FlowJo's and FlowState's counts side by side.

07 · OPTIONAL AI

An AI assistant, only if you want one

Off by default. Each action shows exactly what will be sent first; event data never leave the browser.

- AI auto-analysis: reads what you measured and suggests groups, the setup and the next steps
- Gate with AI: the AI points to the population, the auto gate draws the boundary
- Gating-strategy suggestions and marker labels from your panel
- Drafts of methods text and figure legends; a check of your analysis for common problems
- Runs on your claude.ai account, your own API key or a lab proxy



Off. Nothing is sent.

AI assistant settings

The AI features are off (default). Nothing is ever sent.

☑ Allow sending plot images

☑ Hide sample and file names from the AI (Sample 1, 2, ...)

Privacy. Something is sent only when you start an AI action, and each action shows exactly what it sends first: rendered plot images, channel names and marker labels, gate names, event counts and statistics, and text you type. Never the event data themselves, the FCS files or their keywords (except the instrument name).

Requests go to Anthropic: in claude.ai under your own Claude account, otherwise under the account of the API key (directly or through your lab's proxy), and are handled under that account's terms. Do not use the AI features with identifiable patient data unless your institution permits it. The key is kept in memory only (unless you tick Remember) and is never written into workspaces, exports or logs.

Cost. In claude.ai, requests count toward your plan's usage. With an API key or proxy they are billed per token: a gate request (one plot image and its context) costs well under 1 US cent with a Sonnet-class model, a methods text or an analysis check a few cents; a strategy run makes one request per step.

AI answers can be wrong. Gates are shown as previews you accept or cancel, texts are drafts to check.

AI tools in the ribbon, and the settings with the privacy statement.

Your data stay with you

Everything runs in your browser. Nothing is uploaded, and the app contacts no server unless you switch on the AI assistant.

Save and share

Autosave keeps your session in this browser. Save the workspace, or save it with data as one .zip, to continue on another computer or hand it to a colleague.

Good to know

Beta software: unexpected behaviour can occur, so verify critical results. For research use only, not for diagnostics. The ? button opens Getting started and all shortcuts.

FlowState Free browser-based flow cytometry analysis · AGPL-3.0

Questions and feedback: Dr. Rüdiger Groß, Institute of Molecular Virology, Ulm University Medical Center · flow@nanoscale.app